RCED Model v4.2.2

Network of Democratic Economic Cells

Corrected, Validated and Optimised Version for Switzerland

A Decentralised, Transparent and Equitable Economic Framework for Switzerland

Reseau de Cellules Economiques Democratiques | Netzwerk Demokratischer Wirtschaftszellen

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Transparency

Blockchain & democratic governance

Equity

Progressive taxation & redistribution

Resilience

Cooperatives & citizen sovereign fund

Key Results - Reference Scenario - Year 20

	72	0.185		19.2%
GDP Year 20	GNH / 100	Gini 2046	Hours / week	Fund Surplus

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1. Executive Summary

The RCED (Reseau de Cellules Economiques Democratiques / Netzwerk Demokratischer Wirtschaftszellen — Network of Democratic Economic Cells) proposes a radical yet pragmatic transformation of the Swiss economic system. Version **v4.2.2** corrects the critical errors of v4.2.1 while preserving the model's founding principles: transparency (blockchain), progressive taxation, cooperative governance, and democratic control.

This version introduces major improvements:

- **Mathematically corrected gain-sharing mechanism:** alpha = 30-60% (working time reduction), beta = 30-50% (sovereign fund), gamma = 10-30% (reallocation). Strict constraint: alpha + beta + gamma = 1.
- **Institutional reallocation system:** official Swiss training pathways (Federal Vocational Certificates, Federal Diplomas, IFFP) replace precarious alternative models.
- **Financial optimisation via wage adjustment:** salaries realigned to FSO medians; the differential D_wages(t) is directly reinvested into the Sovereign Fund.

Key Results — Year 20 — Reference Scenario

275	72/100	0.185	28h	19.2%
GDP Index (2026=100)	Gross National Happiness	Gini Coefficient	Hours per week	Sovereign Fund Surplus
+175%	+14 pts	-0.135	-13h	

The model demonstrates that it is possible to simultaneously:

- Reduce working time to **28 hours per week** without any loss of income.
- Grow GDP by **175%** through productivity gains and cooperative efficiency.
- Reduce inequality to a **Gini of 0.185** (below Switzerland's current 0.320).
- Maintain a robust surplus (**>19% of GDP** at maturity) to fund network propagation.

2. Corrections Applied to RCED v4.2.1

Version v4.2.2 corrects the inconsistencies of the previous version to ensure absolute viability within the Swiss institutional context:

Issue in v4.2.1	Correction in v4.2.2	Justification / Impact
Incorrect mathematical formulas (e.g. alpha + beta + gamma != 1)	Strict formulation: alpha + beta + gamma = 1	Restores accounting coherence: 100% of gains allocated, no monetary creation.
Unrealistic initial wages (e.g. CHF 6,500 for a factory worker)	Realigned to FSO medians (e.g. CHF 5,200). Differential is reallocated.	Macroeconomic credibility. D_wages(t) boosts the Sovereign Fund from year 1.
Unrecognised training (P2P EducationLibre) and excessively short durations	Replaced by Federal Vocational Certificates, Federal Diplomas, IFFP. Duration: 12-24 months.	Ensures legal recognition to practise in healthcare and teaching.
Sovereign Fund equation not cumulative in the main text	S(t) = S(t-1) + beta x Delta_P(t) x GDP(t) x fiscal_redistribution x 0.35 + D_wages(t)	Enables real calculation of capital accumulation for propagation.

3. The Corrected RCED v4.2.2 Model

3.1 Core Mechanisms

The model operates through **10 interconnected layers**. Version v4.2.2 replaces P2P education with institutions recognised by SEFRI and the Swiss cantons:

Layer	Element	Function
0	Diagnostic	Piketty / Saez / Zucman - Document inequalities and tax evasion
1	Infrastructure	Global blockchain - Makes wealth visible and taxable
2	Capture	Inverted cantonal taxation - 40 / 35 / 25 key
3	Capitalisation	Citizen sovereign fund - Invests in the real economy
4a	Production	Worker cooperatives - What we produce and how
4b	Consumption	Consumer cooperatives - Democratic supply planning
5	Health / Time	Solidarity and Cooperative Health Insurance + reduced week - Cooperative health, inclusion, free time
6	Coordination	Democratic coordination - Federated cells, market as signal
7	Governance	Swiss model - Subsidiarity, from communes to the Confederation
8	Knowledge	Official Training (SEFRI / OrTra) - Institutional diploma-granting retraining
9	Propagation	0% interest investment - Creates cells elsewhere
10	Time	Work indexation - Working hours fall with automation

Productivity Gain Sharing Mechanism (Mathematically Corrected)

For each year t:

```
1.  Productivity gain:  Delta_P(t) = (1 + automation_rate)^t - 1

2.  Allocation (absolute constraint: alpha + beta + gamma = 1):
      alpha x Delta_P(t)  ->  Working time reduction
      beta  x Delta_P(t)  ->  Sovereign fund capitalisation
      gamma x Delta_P(t)  ->  Sectoral reallocation (Training)
```

3.2 Sectoral Reallocation System

Switzerland faces structural labour imbalances. Unlike previous versions, training is delivered through institutions recognised by **SEFRI** and the cantons, guaranteeing the value of qualifications on the labour market:

Original Job	Target Job	Training Structure	Duration	Success Rate	Revised Salary (CHF/month)
Industrial worker	Hospital maintenance technician	Federal Diploma - Maintenance Technician (vocational schools)	12-24 months	78%	5,200 -> 6,800
Cashier	Healthcare and Community Care Assistant	Federal Vocational Certificate in Healthcare and Community Care or official Red Cross certification	12-24 months	82%	4,100 -> 5,100

Original Job	Target Job	Training Structure	Duration	Success Rate	Revised Salary (CHF/month)
Truck driver	Medical supplies deliverer	Hazardous materials transport certification / medical logistics	2 months	95%	4,800 -> 5,400
Carpenter	Craft education teacher	Pedagogical diploma for vocational teaching (IFFP / HEFP)	12 months	80%	5,500 -> 6,500
Farmer	Therapeutic gardener	Post-vocational specialisation in horticulture / care (OrTra)	6 months	85%	4,200 -> 4,900

3.3 Phased Implementation

Non-negotiable sequencing principle: each phase must prove its value before the next can begin. The surplus must exceed 2% of GDP before any propagation.

Phase	Horizon	Key Actions	Indicators (end)	Surplus (% GDP)
1: Pilot cooperative	Years 0-4	Launch production + consumption cooperative. Test 40/35/25 fiscal key. Integrate wage adjustments.	GDP: 122 | Hours: 40h	0.8% (Boosted via D_wages)
2: Cantonal pilot	Years 4-10	Expand to 50% of canton. Deploy partnerships with public vocational schools.	GDP: 166 | Hours: 37h	2.8%
3: Propagation begins	Years 10-15	First 0% interest investment to neighbouring canton. Cross-border coordination.	GDP: 213 | Hours: 35h	5.5%
4: International network	Years 15-20	Expand to 50% of cantons. Full implementation of 28h working week.	GDP: 275 | Hours: 28h	19.2%

4. Scenarios and Results

Three scenarios are modelled to demonstrate the robustness of the model. Results at Year 20:

Indicator	Conservative	Reference	Optimistic	Baseline 2026
GDP Index (2026=100)	197	275	369	100
GNH Score (/100)	63	72	74	58
Gini Coefficient	0.250	0.185	0.180	0.320
Hours / week	40	28	23	41
Surplus (% GDP)	9.2%	19.2%	23.5%	0%
Active cells	15	20	25	0

- Even the **conservative** scenario achieves significant improvements: GDP +97%, surplus 9.2%.
- The **reference** scenario balances all objectives: strong growth, reduced inequality, viable propagation.
- The **optimistic** scenario maximises all dimensions but requires very high adoption of training pathways.
- All scenarios maintain a surplus **>2% of GDP**, the necessary condition for propagation.

5. Key Assumptions

Category	Assumptions
Economic	Annual automation rate: 2 to 6% (OECD). Keynesian multiplier: 1.5. Productivity gains captured by cooperative structures. Offshore capital repatriation: 60% over 10 years.
Social	Adoption of official training pathways: 25 to 80% depending on scenario. Cooperative coverage: 20 to 65%. Worker acceptance of retraining: 70%. Acceptance of working time reduction: 80%.
Political	40/35/25 fiscal key implemented at cantonal level. 0% propagation requires surplus >2% of GDP. Democratic governance of the sovereign fund.

- These assumptions are calibrated to be internally consistent with the model's logic.
- They are plausible based on existing literature and precedents (Mondragon, Raiffeisen).
- They have not been empirically validated for RCED at this scale — pilot tests are required.

6. Limitations and Caveats

No model is perfect. RCED v4.2.2 honestly acknowledges its limitations:

Political Feasibility

Requires constitutional changes in Switzerland (40/35/25 fiscal key). Potential resistance from net contributing cantons.

Economic Assumptions

An automation rate of 4%/year may be optimistic in the long run for service sectors (Baumol's cost disease).

Implementation Risks

The longer institutional training durations (12 to 24 months) create temporal inertia in sectoral reallocation. Federal academic requirements may slow rapid workforce integration.

Data Limitations

Swiss wealth data (SNB 2026) may underestimate actual offshore holdings.

To address these limitations:

- Test in 1-2 pilot cantons (Geneva, Vaud) before national rollout.
- Independently validate key assumptions with economic experts.
- Implement adaptively with transparent reporting of both successes and challenges.

7. Sources and References

Theoretical Foundations

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- Saez, E., & Zucman, G. (2019). The Triumph of Injustice
- Keynes, J.M. (1930). Economic Possibilities for our Grandchildren
- Hayek, F. (1945). The Use of Knowledge in Society

Swiss Data

- SECO (2026). Swiss GDP
- SNB (2026). Total national wealth
- FSO (2026). Swiss wage medians and population
- SEFRI - State Secretariat for Education, Research and Innovation
- OrTra - Occupational organisations

Reference Websites

- www.bfs.admin.ch
- www.seco.admin.ch
- www.snb.ch
- www.bag.admin.ch
- www.oecd.org
- github.com/nadnone/EducationLibre (CC BY 4.0)

8. Appendices

8.1 Glossary

Term	Definition
Federal Vocational Certificate (CFC)	Swiss vocational qualification awarded after 2-4 years of apprenticeship training
RCED	Reseau de Cellules Economiques Democratiques / Network of Democratic Economic Cells
ESS	Economie Sociale et Solidaire / Social and Solidarity Economy
Healthcare and Community Care Assistant	Swiss-recognised vocational qualification for community-based healthcare support
Gini	Gini coefficient: inequality index (0 = perfect equality, 1 = maximum inequality)
GNH	Gross National Happiness (composite wellbeing indicator)
SEFRI	State Secretariat for Education, Research and Innovation (Switzerland)
OrTra	Occupational organisations responsible for vocational training frameworks
40/35/25 Key	Fiscal distribution: 40% cantons, 35% sovereign fund, 25% equalisation
alpha / beta / gamma	Productivity gain allocation parameters: time reduction / sovereign fund / reallocation (sum = 1)

8.2 Mathematical Formulas

Core formulas used in the RCED v4.2.2 simulation:

1. Allocation Constraint

```
alpha + beta + gamma = 1
```

2. Productivity Gain

```
Delta_P(t) = (1 + r)^t - 1
(r = automation rate, 2 to 6% depending on scenario)
```

3. Working Time Reduction

```
H(t) = H(t-1) x (1 - alpha x Delta_P(t))
```

4. Cumulative Sovereign Fund Surplus (Corrected v4.2.2)

```
S(t) = S(t-1)
      + beta x Delta_P(t) x GDP(t) x fiscal_redistribution x 0.35
      + D_wages(t)
(D_wages = differential between FSO median wages and previous wages)
```

5. Sectoral Reallocation

```
R(t) = gamma x Delta_P(t) x L(t)
```

8.3 Sensitivity Analysis

The integration of the classical training system considerably stabilises the model. Although retraining (gamma) takes longer (up to 24 months), the retention rate in new jobs increases structurally thanks to the value of state-recognised qualifications under Collective Labour Agreements (CCT). The beta parameter (Sovereign Fund) benefits massively from the addition of D_wages(t), making propagation financially viable from the very first phase.

Assumption	Base Case	Test Case	GDP Impact (Y20)	Gini Impact (Y20)
Automation rate	4%/yr	2%/yr	-50 pts	+0.02
Automation rate	4%/yr	6%/yr	+100 pts	-0.01
Cooperative coverage	45%	20%	-80 pts	+0.04
Cooperative coverage	45%	65%	+90 pts	-0.02
Fiscal redistribution	28%	20%	-40 pts	+0.03
Fiscal redistribution	28%	35%	+50 pts	-0.02
alpha (time reduction)	50%	30%	+20 pts	0.00
beta (Sovereign Fund)	40%	30%	-15 pts	+0.01
gamma (reallocation)	10%	20%	-10 pts	+0.01